House Price Prediction System

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TABLE OF CONTENTS

1. **Problem Description 3**
   - **1.1. Dataset Description 3-4**
   - **1.2. Flow diagram 5**
2. **Modules for Project Implementation……..……6**
   - **2.1. Modules Used 6-7**
   - **2.2. Platform Used 8**
3. **Machine Learning model/ Searching Technique used 9**
4. **Evaluation metrics used 10**
5. **Results and Visualizations 11-12**
6. **Conclusion and Future Scope 13**
7. **References..... 14**

1. Problem Description

The House Price Prediction System is a machine learning-based application designed to estimate property prices based on various features such as location, size, number of rooms, and other relevant factors. In the real estate market, accurately predicting house prices is a complex task due to multiple influencing variables. This project aims to automate the prediction process using machine learning techniques, reducing human effort and improving decision-making for buyers, sellers, and investors. The system uses data preprocessing, feature selection, and regression models to learn patterns from historical housing data and generate accurate price predictions. This helps in understanding market trends and making informed financial decisions.

1.1 Dataset Description

The dataset used in this project contains multiple features that influence house prices. These features include both numerical and categorical variables that describe the physical and locational characteristics of a property.

1. Structural & Core Property Features : These features describe the physical attributes of the house itself.

- **property_area_sqft**: The total living area of the property. This is the most important feature .
- **bedrooms**: Total number of bedrooms.
- **bathrooms**: Total number of bathrooms.
- **floors**: The number of levels/stories in the building.
- **year_built**: The year the property was originally constructed.
- **property_age**: Calculated as (Current Year - Year Built). Newer houses typically hold different value profiles than historic ones.
- **property_type**: Categorized into Apartment, Independent House, or Villa. (Processed via One-Hot Encoding in your model).
- **lot_size_sqft**: The total area of the land plot the house sits on.

2. Location & Connectivity : Features that define the property's relationship to the surrounding city.

- **distance_city_center_km**: How far the property is from the central business district. It has a negative correlation with price (prices drop as distance increases).
- **public_transport_score**: Accessibility to buses, trains, or subways (Scale 1–10).
- **shopping_distance_km**: Proximity to the nearest major malls or markets.
- **hospital_distance_km**: Proximity to healthcare facilities.

3. **Neighborhood & Quality of Life** : Socio-economic and environmental factors of the area.

- **neighborhood_quality**: A general score (1–10) representing the prestige or upkeep of the area.

- **crime_rate_index**: Safety level of the neighborhood (Lower usually means higher property value).
- **school_rating**: The quality of local educational institutions.
- **green_space_index**: Measure of nearby parks and nature (Scale 1–10).
- **noise_pollution_level**: Ambient noise levels (Scale 1–10).

4. Technical & Utility Scores : Internal quality and infrastructure reliability.

- **construction_quality**: The grade of materials and workmanship used (Scale 1–10).
- **energy_efficiency**: How well the house retains heat/cooling (Scale 1–10).
- **water_supply_score**: Reliability of water services.
- **electricity_supply_score**: Reliability of the power grid.
- **internet_score**: Quality of high-speed connectivity available at the address.

5. Status & Hazards

- **renovation_status**: Binary (0 or 1) indicating if the property has been recently updated.
- **parking_available**: Binary indicating if the property includes dedicated parking.
- **flood_risk_index**: The likelihood of water damage based on geographical data (Scale 1–10).

6. Target Variable

- **house_price**: The market value of the home. This is what your Linear Regression and Random Forest models are attempting to predict.

1.2 Flow Diagram

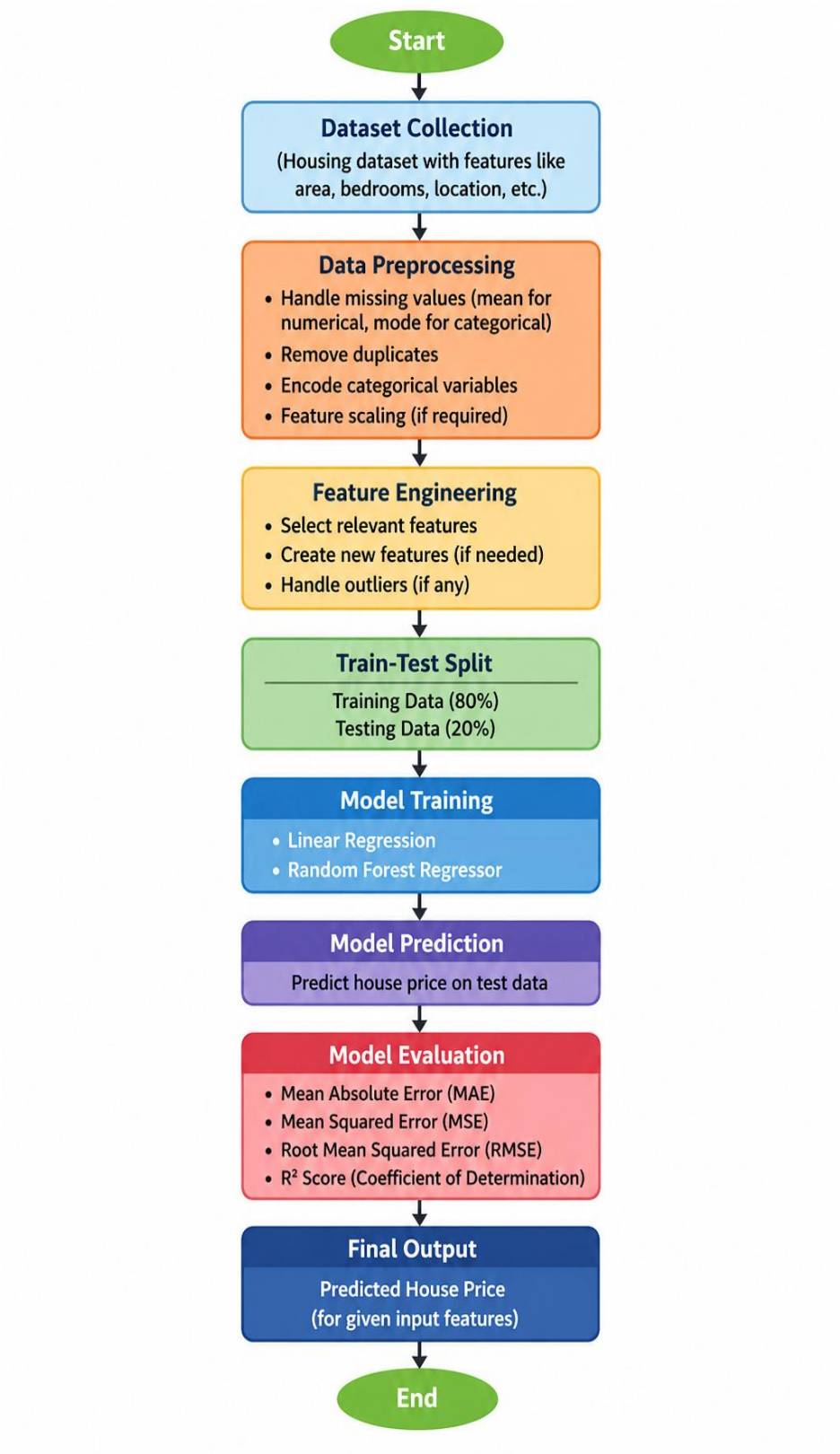

Fig.1 Sample flow diagram

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

1. Pandas

Pandas is a powerful library used for data manipulation and analysis.. It provides data structures like DataFrames which make it easy to handle structured data. Used for data cleaning, preprocessing, and organizing the dataset.

Functions Used:

- **pd.read_csv()** : Used to load the dataset from a CSV file into a DataFrame
- **df.head()** : Displays the first few rows of the dataset
- **df.info()** : Provides summary of dataset (data types, null values)
- **df.describe()** : Generates statistical summary
- **df.drop()** : Removes unnecessary columns
- **df.fillna()** : Handles missing values
- **df.select_dtypes()** : Separates numerical and categorical columns

2. NumPy

NumPy is used for numerical computations and handling arrays efficiently. Used for mathematical operations and efficient numerical processing

Functions Used:

- **np.sqrt()** : Used to calculate RMSE from MSE
- **np.array()** : Converts data into array format

3. Scikit-learn (sklearn)

Scikit-learn is a machine learning library used for building models, preprocessing data, and evaluating performance. Core library for implementing machine learning pipeline including training, testing, and evaluation.

Functions Used:

- **train_test_split()** : Splits dataset into training and testing sets (e.g., 80% training, 20% testing)
- **SimpleImputer()** : Used to handle missing values :-
    - Mean strategy → for numerical columns
    - Most frequent → for categorical columns
- **LinearRegression()** : Used for predicting house prices based on linear relationships
- **RandomForestRegressor()** : Ensemble model that improves prediction accuracy using multiple decision trees
- **mean_absolute_error()** : Calculates MAE
- **mean_squared_error()** : Calculates MSE
- **r2_score()** : Calculates R^2 Score

4. Matplotlib

Matplotlib is used for creating visualizations and plots. Used to visualize model predictions and trends in data.

Functions Used:

- **plt.plot()** : Line plots
- **plt.scatter()** : Scatter plots
- **plt.show()** : Displays graphs

5. Seaborn

Seaborn is a visualization library built on top of Matplotlib, used for creating more attractive and informative statistical graphics.

Functions Used:

- **sns.heatmap()** : Used to display correlation matrix

2.2 Platform Used: (mention all the platforms with the reason of using it)

- **Visual Studio Code**: For code writing, testing, and debugging.
  VS Code is equipped with multiple extensions. Provides debugging tools for error detection. Supports multiple programming languages including Python. Easy file management and project organization

- **Jupyter**: Allows combining code, output, and explanations in one place, Ideal for data analysis and visualization, Helps in debugging and testing models easily, Provides clear output for graphs and tables.

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

1. Linear Regression

Linear Regression is a supervised learning algorithm used for predicting continuous values such as house prices. It assumes a linear relationship between independent variables (features) and the dependent variable (price).

Working:
The model finds the best-fit line that minimizes the difference between actual and predicted values.

Advantages:
- Simple and easy to implement
- Fast computation
- Easy to interpret results

Limitation:
- Cannot capture complex non-linear relationships

2. Random Forest Regressor

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs to improve accuracy.

Working:
- Multiple decision trees are created using random subsets of data
- Predictions from all trees are averaged

Advantages:
- Handles non-linear relationships effectively
- Reduces overfitting
- Provides higher accuracy in complex datasets

Limitation:
- Slower compared to Linear Regression
- Less interpretable

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

The performance of the models is evaluated using the following metrics:

1. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

Measures the average absolute difference between actual and predicted values. Lower MAE indicates better model performance.

2. Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Gives more weight to larger errors, making it useful for identifying models with large deviations.

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

Provides error in the same unit as the target variable (price), making it easy to interpret.

4. R^2 Score (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Represents how well the model explains the variance in the data. A value closer to 1 indicates a better fit.

5. Results and Visualizations

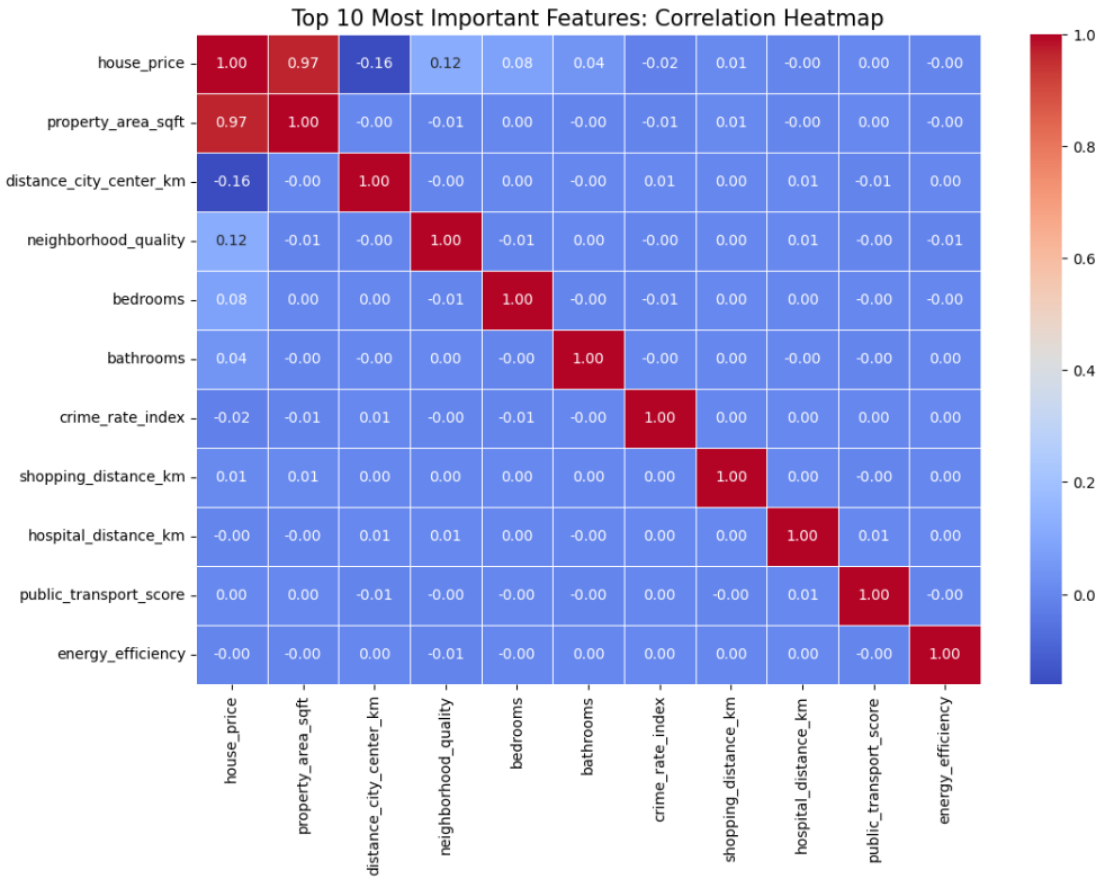

Fig.2 Heatmap showing corelation.

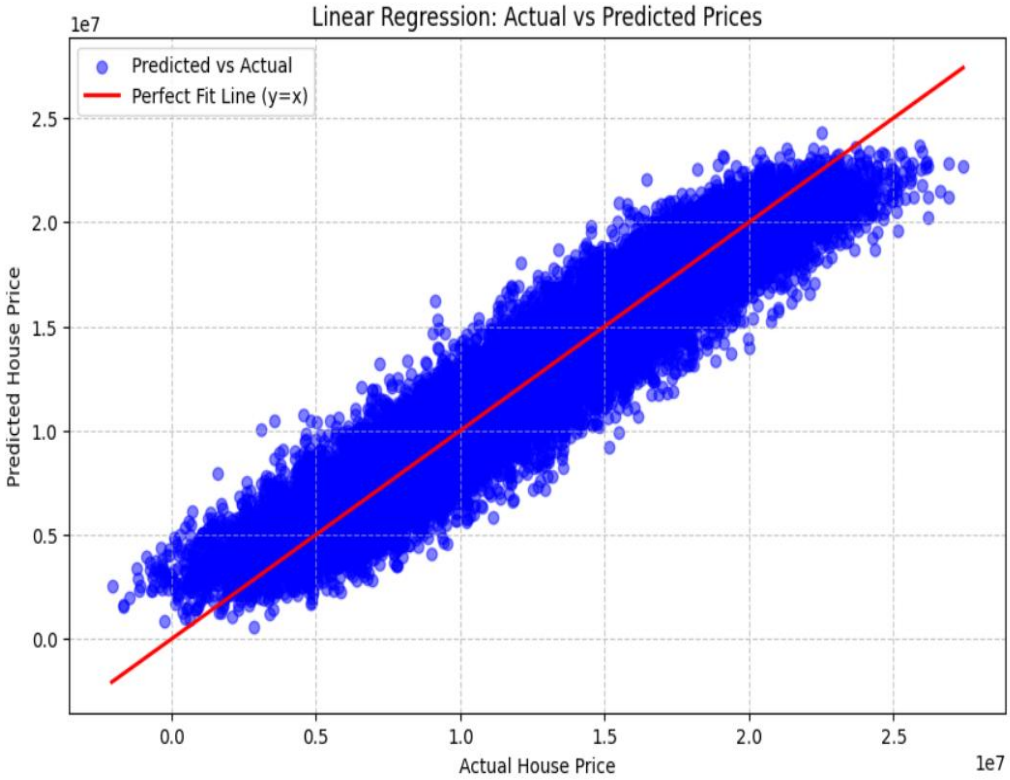

Fig.3 Linear Regression Best Line Graph

6. Conclusion :

The House Price Prediction System was successfully developed using machine learning techniques. The models were trained and evaluated using multiple performance metrics.

- Linear Regression showed strong performance with high R^2 score
- Random Forest handled complex patterns but had slightly higher error

Overall, the system is capable of predicting house prices with good accuracy.

Future Scope:

- Use larger and real-time datasets
- Apply advanced algorithms like XGBoost or Gradient Boosting
- Deploy the model as a web application
- Integrate user interface for real-time prediction
- Improve accuracy using hyperparameter tuning

7. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – https://www.geeksforgeeks.org/
- TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
- Stack Overflow – https://stackoverflow.com
- Scikit-learn Documentation -- https://scikit-learn.org
- Chatgpt -- https://chatgpt.com/c/69e66a5a-6708-8324-97f1-428e1440bb70, https://chatgpt.com/share/69f0bc3f-70a8-83e8-8ef6-56dab4e802ad